

SUMMARY

Geophysical Engineering graduate with hands-on experience in document and data extraction, file format conversion (PDF, XLSX, CSV, LAS), and data labeling or classification for both geoscience and non-geoscience datasets. Proficient in using Python for data management, automation, and structured data processing. Strong attention to detail proven through large-scale data validation (more than 10,000 entries) and collaborative projects. Keen to apply technology-driven methods for document classification and data digitization in oil & gas industry.

EDUCATION LEVEL

INSTITUT TEKNOLOGI SEPULUH NOPEMBER (Sep 2020 – Apr 2025)

Bachelor of Geophysical Engineering - GPA 3.41

- Bachelor Thesis Title: "Performance Enhancement Of XGBoost Using Continuous Restricted Boltzmann Machine And Particle Swarm Optimization Method In Poseidon Field, Browse Basin, Australia For Lithology Prediction".
- Relevant Coursework: Geophysical Computing, Mathematical Geophysics, Geostatistics, Well Log Data Analysis, Soft Computing.

WORK EXPERIENCE

BADAN PUSAT STATISTIK (CENTRAL BUREAU OF STATISTICS)

Data Analyst Intern (Jan 2024 – Apr 2024)

- Validated and maintained more than 10,000 structured data entries for Daerah Dalam Angka (DDA) reports, ensuring data accuracy and consistency.
- Extracted and digitized tabular data from PDF reports into structured CSV/XLSX formats, then transformed into a poster-style data visualization dashboard.
- Executed large-scale data cleaning, integrity checks, and structured reporting to support regional planning decisions.
- Built and deployed a WebGIS platform to improve geographic data accessibility for more than 200 stakeholders.

PROJECT EXPERIENCE

DATA ANALYSIS PROJECT (EKLIPSE.GG RECRUITMENT ASSESSMENT)

Data Analyst Candidate (Nov 2025 – Dec 2025)

- Successfully completed the technical recruitment assessment for the Junior Data Analyst position, resulting in a job offer.
- Integrated and analyzed multi-table user activity datasets using Python (Pandas) and SQL (SQLite).
- Developed analytical SQL queries to evaluate user engagement, premium conversion behavior, gameplay activity trends, and content creation patterns.
- Generated actionable insights through data visualization and exploratory analysis to support product and engagement strategy recommendations.
- Performed end-to-end workflow including data cleaning, database querying, aggregation, and reporting.

PETROPHYSICS SOFTWARE DEVELOPMENT IN PETROPHYSICS LAB ITS

Software Developer (Jan 2025 – May 2025)

- Developed academic-scale petrophysics software inspired by industry tools (e.g., Interactive Petrophysics).
- Designed and implemented log visualization and calculation modules (porosity, water saturation).
- Constructed data extraction logic from structured tables and GUI for depth based well log interaction.
- Structured log input handling and computational logic for multi-curve datasets.
- Presented the software at IPA Convex 2025 Youth Project Booth.

LITHOLOGY MODELING PROJECT IN SCALE UP GEOENERGY HACKATHON 2025

Data Scientist - Well Log Data & Modeling (Feb 2025 – May 2025)

- Built an auto-labelling system for lithology classification using machine learning, applying Continuous Restricted Boltzmann Machine (CRBM), Particle Swarm Optimization (PSO), and Bayesian Optimization to enhance XGBoost performance.
- Processed and managed ~22 multi-well LAS datasets for lithology classification.
- Conducted log preprocessing including missing interval handling, ML-based log imputation and distribution normalization.
- Applied spatially clustered modeling based on well proximity to reduce geological heterogeneity.
- Performed well-level data splitting to prevent data leakage.
- Successfully performed automated lithology labelling on blind test wells (F1 score: 60%) – demonstrating understanding of data labeling workflows, quality control, and model-driven classification.

INTEGRATED FIELD COURSE PROJECT BROMO TENGGER SEMERU

Jr. Geophysicist (Field Project) (May 2023 – Dec 2023)

- Conducted multi-method geophysical data acquisition (1D/2D geoelectrics, seismic, MT, GPR, magnetic, gravity, aeromagnetic, aeroGPR), collecting and processing 500+ data points under structured field procedures.
- Led microtremor/passive seismic data processing, including preprocessing, noise filtering, and initial interpretation, improving data accuracy by ~30% and generating reliable subsurface models.
- Performed geological mapping and integrated geophysical datasets across 10+ observation points to produce subsurface interpretations, including identification of structures, lithology variations, and potential resource indicators.

ACHIEVEMENTS

CHAMPION OF OIL AND GAS SMART COMPETITION PROTECT 2024 BY SPE UP

Team Leader – (May 2024)

- Secured victory against 20+ teams from 8+ top universities in a smart competition covering upstream to downstream oil & gas knowledge, statistics, and trivia.
- As the team leader, I successfully mentored and guided my team (batch 2022 and 2023 students), leveraging their full potential to achieve success.
- Served as the team's key contributor and subject matter expert in Geosciences, Drilling, and Production topics, including well completion, well workover and intervention, petrophysics, formation evaluation, PVT data fundamentals.

1ST PLACE WINNER OF SEG CHALLENGE BOWL APRQ 2022

Team Leader – (Oct 2022)

- Achieved a dominant victory against over 20 teams from across the Asia-Pacific region, including Indonesia, Australia, Malaysia, China, Philippines, and India in a smart competition covering geoscience, geodesy, and astronomy knowledge, statistics, and trivia.
- Secured a commanding win with a final score of over 400 points, establishing a significant gap of more than 300 points over the second and third-place teams (Indonesia and India).
- Provided key expertise and knowledge in Geology, Geophysics, Exploration Seismic, and Astronomy topics, which were crucial to the team's success.

ORGANIZATIONAL EXPERIENCE

GEOPHYSICAL ENGINEERING STUDENT ASSOCIATION (HMTG)

Expert Staff in Professionalism and Science Division (Feb 2022 - Dec 2023)

- Oversaw student activities in geophysical engineering in scientific fields such as Pekan Kreativitas Mahasiswa (PKM) and competition communities (Geoscience Club).
- Simultaneously contributed to event branding and design, creating and producing key materials such as ID cards, banners, and promotional content to ensure a professional and cohesive visual identity for all division events.

SOCIETY OF EXPLORATION GEOPHYSICIST (SEG) ITS SC

Officer in Human Resource Development Division (Feb 2022 - Mar 2023)

- Designed and facilitated training programs and skill-upgrading sessions to enhance members expertise in exploration geosciences.
- Managed member data and ensured effective communication to keep all members informed about key events and internal activities.

SKILLS

Technical Skills: Data Extraction & Digitization, Multi-format Data Handling (LAS, CSV, Excel, PDF), Data Labeling & Classification, Data Cleaning & Validation, Machine Learning & Data Analytics, Geoscience Operations Fundamentals, Drilling Operations Fundamentals, Petrel, Landmark Halliburton, Dataiku
Programming Skills: Python (Pandas, NumPy, Scikit-learn, Matplotlib), SQL (SQLite)
Soft Skills: Attention to Detail, Analytical Thinking, Team Collaboration, Effective Communication, Leadership